Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[14EE35]**

THIRD SEMESTER B.TECH. EXAMINATION, DECEMBER-2015

DIGITAL ELECTRONICS CIRCUITS

Time:3 Hours **Maximum Marks:70**

Instructions : Answer five full questions, choosing at least TWO from each part.

PART - A

1. a. Simplicity the given function and draw the logic diagram using basic gates. **06**

$f(a,b,c) = \bar{a}\,\bar{b}\,\bar{c} + \bar{a}\,\bar{b}\,c + a\,\bar{b}\,\bar{c} + abc$

b. Represent the following in both canonical maxterm and minterm forms in decimal notation **04**

$P = f(a,b,c) = a\bar{b} + a\bar{c} + bc\ (SOP)$

$Q = f(w,x,y,z) = (\bar{w}+x)\ (y+\bar{z})(POS)\cdots\cdots\cdots$

c. Determine the dual of the Boolean expression: **04**

i. $AB + \bar{A}BC + BC = \bar{B}$

ii. $(b+\bar{c})(\bar{b}+c)(\bar{a}+c) = \bar{a}\bar{b}\bar{c}$

2. a. For a given function find the prime implicants and essential prime implicants also find the minterm expression. Realize the final expression using only NAND gate. **07**

$F_1 = f(a,b,c,d) = \Sigma(1,5,7,8,9,10,11) + d(13,15)$

b. A 4 bit binary character is presented a circuit that must detect whether the input is a legitimate BCD code. If non-BCD code is entered, the output is to be true (logic 1). Construct the truth-table. Write Boolean output equation. **07**

3. a. Find the minimal sum using Quine-McCluskey method and table-reduction method. **08**

$f(a,b,c,d) = \Sigma(0,1,2,3,6,7,8,9,14,15)$

b. Reduce the given function using map-entered variable. **06**

$f(a,b,c,d) = \Sigma(2,3,4,5,13,15) + \Sigma d(8,9,10,11)$

4. a. Implement the multiple function using two 3:8 decoder: **06**

$f_1(w,x,y,z) = \Sigma(3,7,9,13)$

$f_2(w,x,y,z) = \Sigma(0,4,8,10,14,15)$

b. Design a priority encoder for a system with 3 input with the middle bit with highest priority encoding to 10, the MSB with the next priority encoding to 1 1 while the LSB with the least priority encoding to 01. **08**

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PART - B

5. a. Realize the given function using 8:1 Mux. (74151) **07**

$T = f(w,x,y,z) = \Sigma(0,1,2,4,5,7,8,9,12,13)$

b. Design 2 bit comparator sketch block diagram. Construct truth-table and simplify output equation and draw the resulting logic diagram. **07**

6. a. Explain the working of master-slave JK flip-flop with the help of logic diagram, function table and timing diagram. **07**

b. Explain the functioning of positive edge triggered D-flip-flop with timing waveforms. **07**

7. a. Derive characteristics equation of SR and JK flip-flop. **06**

b. With neat diagram and with all modes of operation design universal shift register using positive edge triggered D-flip-flop. **08**

8. a. Design synchronous mod-6 counterusing clocked SR flip-flop, for a sequence 0-2-3-6-5-1. **10**

b. Explain the operation of twisted ring counter with all states.(mod -7 twisted ring counter). **04**